Xnergy

Autonomous Power Technologies

ORIGIN FOR (SG) WESTON ROBOT

Application Note

BEAB5F2– Troubleshoot Guide for Beyond Series

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Abstract

Xnergy BE series wireless charger incorporates wireless power transfer technology to charge battery through medium of air without physical contact. Wireless technology is widely used in small power mobile devices, and Xnergy is the pioneer to offer industrial-grade high power wireless solution. Various interfaces are provided for a better integration, but at the same time it also increases the chances of having difficulties. This document provides step by step guidance to locate the root cause of unintended behaviours and provide solutions to rectify issues.

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1. Introduction

Xnergy BE series wireless charger incorporates wireless power transfer technology from transmitter to receiver and charge batteries. The power transfer process is done without physical contact unlike conventional wired charger. Xnergy is a pioneer of providing industrial-grade high power wireless charging solutions. User Manual and Application Notes are provided for the users to understand how to operate and control Xnergy BE series wireless charger. Follow those technical documentations will minimize the integration effort, however due to the complexity of modern robot system, sometimes debug is unavoidable. This document supplements the User Manual and Application Notes by providing a step-by-step trouble shoot guide to identity problems and issues.

A typical Xnergy BE series charger consists of three parts as shown in below figure.

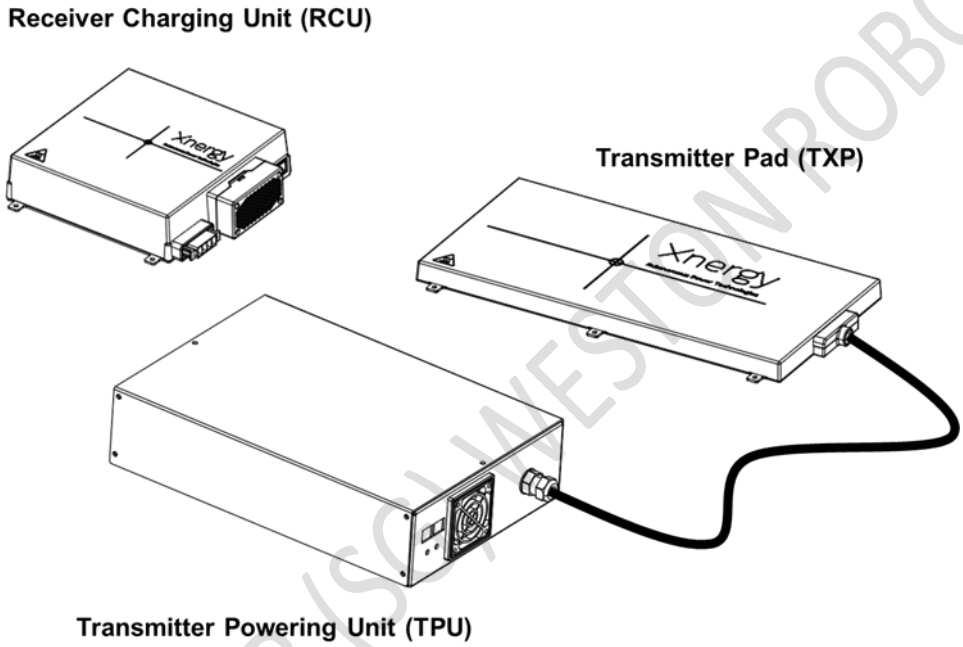

Figure 1 type Xnergy BE series charger

2. Check TPU and RCU condition.

To ensure safety, user is required to take the following prerequisite steps before trouble shooting.

1. Stop charging. This can be done by sending signal from the RCU or worst case, just cut-off the input power to TPU.
2. TPU/TXP and RCU must be placed away from each other at least 10cm.

After stopping charging and put TPU and RCU away, we can proceed to check the condition of the charger. Please follow the steps below:

1. Visual check
   Please inspect TPU, TXP and RCU visually. Check is there any damages on cablings, transmitter pad (TXP), TPU, or RCU. Please ensure that there is no physical damage on any part of the charger. If there is, please contact Xnergy customer service and stop trouble shooting.
2. Check power connection
   Power cables of TPU must be installed properly and the connection should not be loose. Please ensure power line, neutral and earth are connected properly to power socket. For RCU, please

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assemble RCU's receptacle connector based on *BET7E51 – User Manual* and follow the instructions. Reverse polarity may damage RCU. Fire hazard may be present when cable sizes is not selected properly. Please be noted, cable selection is critical for high current application.

3. Check input AC voltage for TPU.
   For TPU, please ensure that the AC input voltage is in between 200 to 250Vac. If the AC input voltage is not within the permitted range, charger may not work properly.
4. Check battery voltage for RCU.
   For RCU, minimum battery voltage is 15V and the maximum battery voltages are 30V and 60V for 24V model and 48V model, respectively.
5. Check fan condition
   Please ensure the fans are not blocked. In Idle mode, Fan of TPU spins slowly and fan of RCU does not spin. If any of the fan is running very fast and significant amount of noise is generated, please disconnect TPU or RCU from the power supply an let it rest for 10~15 minutes, before turning it on. Excessive fan's noise is often caused by over-temperature, please make sure environment is well ventilated and ambient temperature does not exceed 40°C which is maximum ambient operating temperature of the device.
6. Check TPU LED status
   Please power on TPU and monitor the LED status. If red LED is blinking slowly, the TPU is in idle model and waiting for charging request. This means TPU works properly. For details please refer to : *BEAB6BA – Control and Monitor Your Charger.* If both LEDs are OFF or other unknown status and input ac voltage is within the permitted range, please contact Xnergy Service Centre.
7. Check RCU LED status
   Please power on RCU and monitor the LED status. If green LED is blinking slowly, the RCU is in idle model. This means RCU works properly. For details, please refer to: *BEAB6BA – Control. and Monitor Your Charger.* If both LEDs are OFF or other unknown status and battery voltage is within the permitted range, please contact Xnergy Service Centre.

3. Trouble shooting charging

After ensuring charger is in working condition, we can proceed to trouble shoot the charging process. Most of the time, is due to parameters setting or improper usage of the charger. The most common issues are listed below:

1. Fail to start charging.
   a. Please ensure the gap and misalignment between RCU and TXP within the permitted range. Please keep a 5cm airgap between TXP and RCU and make sure the centre of TXP and RCU is aligned and retry charging.
   b. Please check the method to enable charger, which should be one the following: Modbus, CANOpen, IO control and proximity. These methods are mutually exclusive, please do only use one method. For example, if the charger is configured to use IO control but the user tries to use Modbus/XDM to enable charger, this will cause error.
2. Device can start charging but immediately stop charging.
   a. Is the battery already fully charged? When the battery is already fully charged, depends on the setting, charger may stop automatically. Please check the RCU charging profile setting.
   b. Does battery voltage exceed cut-off voltage setting? If the battery voltage already exceeds cut-off voltage, it will stop automatically.

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   c. Does charger's current setting exceed battery's maximum charging current? Please note that, some old batteries have their maximum charging current reduced. Thus, it cannot handle the same current as new battery.
3. Loud noises are generated form the device.
   a. Do users utilise the charger in full power? During full power charging, the charger's temperature may increase. It causes the fan spinning very fast and generating significant amount of noise. It is a normal behaviour. To remove the fan's noise, stop the charging and let it rest for 20-30 minutes to let the charger cool down.
   b. If fan still spin fast, please check whether green LED is blinking and charger still can communicate in Modbus/XDM. If both green and yellow LED are continuously ON or continuously OFF, or communication to XDM is lost, RCU maybe damaged. Please contact Xnergy Service Centre.
4. Device cannot connect to XDM/Modbus.
   a. Do you use correct cable (provided Modbus cable)?
   b. Do you connect to the correct terminal? Only outer side RJ45 of RCU can connect to Modbus as stated in Chapter 2 of *BEAB6BA – Control and Monitor Your Charger*.
   c. Does Modbus have the same setting stated in Table 5 of *BEAB6BA – Control and Monitor Your Charger*?
   d. Do you use proper USB to RS485 converter? Please try the USB to RS485 converter come with the charger.
5. Device cannot communicate with robots/CAN?
   a. Do you use correct cable (provided CAN cable)?
   b. Do you connect to the correct terminal? Both of RJ45 terminal on RCU can connect to CAN BUS.
   c. Does CAN have the same setting stated in Chapter 6 of *BEAB6BA – Control and Monitor Your Charger*?
   d. Does the message follow the format stated in Chapter 6 of *BEAB6BA – Control and Monitor Your Charger*?

4. Error code list of RCU

Below is the error code list of RCU. Based on the error code, issues can be identified. Details are listed in *BEAB6BA – Control and Monitor Your Charger*. Use can try the method list in trouble shoot guide to see if the error can be resolved. If not, please contact Xnergy customer service.

Table 1: RCU Error code description and trouble shoot guide

Error Code High	Error Code Low	Description	Trouble shoot guide
0x0002	0x0000	Foreign object is detected[1]	Remove any metallic items between transmitter and receiver
0x0001	0x0000	Charging initialization error[2]	RCU is not in permitted working gap / alignment and transmitter rejects charging request
0x0000	0x8000	One of fan is stuck	Clean fan
0x0000	0x4000	RCU over temperature detected	Turn off and let RCU rest for 20-30 minutes to cool down

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0x0000	0x0200	Transmitter error during charging	Power off TPU and wait minimum 1 minute and power on
0x0000	0x0100	Communication lost	Communication between transmitter and receiver is lost during charging. please try charging again.
0x0000	0x0002	Transmitter Pairing timeout	Make sure both TPU and RCU have the same pairing ID
0x0000	0x0001	configurations are not set properly	Make sure all setting variables do not exceed 0xFFFF (65535 or 32768)

[1]Foreign object detection (FOD) at RCU is optional. If not equipped, Foreign object detected error will never be reported.

5. Error code list of TPU

Below is the error code list of RCU. Based on the error code, issues can be identified. Details are listed in *BEA957F – Monitor Your Transmitter*. Use can try the method list in trouble shoot guide to see if the error can be resolved. If not, please contact Xnergy customer service.

Table 2:TPU Error code description and trouble shoot guide

Error Code High	Error Code Low	Error Status Message	Trouble shoot guide
0x8000	0x0000	AC input under voltage	Make sure AC input voltage is more than 200Vac
0x4000	0x0000	AC input over voltage	Make sure AC input voltage is less than 250Vac
0x2000	0x0000	AC input over frequency	Make sure AC input voltage is less than 65Hz
0x1000	0x0000	AC input under frequency	Make sure AC input voltage is more than 45Hz
0x0800	0x0000	AC input over current	Internal circuit may have short circuit
0x0400	0x0000	DC bus overvoltage	Make sure AC input voltage is less than 250Vac
0x0200	0x0000	DC bus undervoltage	Make sure AC input voltage is more than 200Vac
0x0008	0x0000	Coil over current	RCU is moving during charging
0x0004	0x0000	Handshake error	Wrong Acceptance ID
0x0000	0x4000	Charger Over Temperature	Turn off and let TPU rest for 20-30 minutes to cool down
0x0000	0x1000	Temperature Sensor 3 Error	Temperature sensor may not be installed properly, or connection come off or damaged. Please contact Xnergy Service Centre to clear the error.
0x0000	0x0800	Temperature Sensor 2 Error	
0x0000	0x0400	Temperature Sensor 1 Error	
0x0000	0x0200	Fan stuck	Clean fan, power off TPU 1 minute and power on. please contact Xnergy Service Centre if error persists.
0x0000	0x0080	Flash memory is not set properly	Make sure all setting variables do not exceed 0xFFFF (65535 or 32768)

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0x0000	0x0040	RF transmitter board is not connected properly	Power off TPU and wait for 5 minutes and power on
0x0000	0x0020	RF Communication lost	Communication between transmitter and receiver is lost during charging. please try charging again.
0x0000	0x0010	Gap between transmitter pad and receiver is too close	Move RCU away from TXP
0x0000	0x0008	Gap between transmitter pad and receiver is too far	Move RCU closer to TXP
0x0000	0x0004	Charging ramp-up timeout	Power cannot be transferred properly, RCU is too far from TXP
0x0000	0x0001	Transmitter fails to boot	TPU cannot start, internal circuit may be damaged. Please contact Xnergy Service Centre to clear the error.

6. Summary

This document provide trouble shooting guide for the most common problems that may happen in the field. Error code lists of both TPU and RCU are given. It is strongly suggested that the user go through the user manual, datasheet, and application notes to understand how the charger works to avoid issues in application.

7. Applicable Firmware revision

List the supported firmware revision of this document, other firmware revision please refer to corresponding documentation.

Firmware Revision	Firmware Version Number Low	Firmware Version Number High
V1.10	0x9FD4	0xED09
V1.11	0x48F8	0xC9A6

8. Revision History

Version Number	Date issued	Author	Update information
V1.0	2020.11.17	Luiz Rizki Ramelan	First published version

9. Important Notice and Disclaimer

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